

Multi-Source Candidate Data Transformer — Technical Design

Eightfold Engineering Intern Assignment · Shruti Sivakumar

Problem Framing

The transformer consolidates noisy, partially overlapping candidate data from multiple sources into a single reliable profile. Since incorrect data is more harmful than missing data in hiring workflows, the design prioritizes **precision, explicit provenance, deterministic merging, and confidence-aware output** over aggressive inference.

Pipeline

```
extract → normalize → resolve identity → merge → score → project → validate
```

`extract` ingests each source independently. `normalize` converts source-specific fields into canonical values with base confidence. `resolve identity` groups records using reliable identifiers. `merge` resolves conflicts and deduplicates data. `score` computes overall profile confidence. `project` reshapes the canonical record using a runtime configuration, and `validate` checks the output against a JSON Schema generated from the same configuration.

The implementation supports structured (Recruiter CSV, ATS JSON) and unstructured (GitHub JSON, recruiter notes) sources through a common adapter interface, making new sources easy to add.

Canonical Schema & Normalization

Each canonical value stores:

```
value, source, method, confidence
```

The canonical profile contains identity, contacts, location, links, headline, experience, skills, education, projects, provenance, and overall confidence.

Normalization includes:

- Phones → E.164 (`phonenumbers`)
- Dates → `YYYY-MM` (`YYYY` preserved where appropriate)
- Countries → ISO-3166 alpha-2
- Emails and URLs → canonical format
- Skills → extracted from text using **n-grams** with a **POS-tagging gate**, then matched using **exact alias lookup** followed by **guarded fuzzy matching** for typo recovery.

Extraction is intentionally conservative—ambiguous tokens such as **R** or **Go** require surrounding **skill** context before being accepted from free text.

Identity Resolution, Merge & Confidence

Identity resolution primarily relies on normalized **email** and **phone**. Name alone never merges records. Unanchored records merge only when the candidate bundle is unambiguous.

Single-value fields select the highest-confidence value using deterministic tie-breaks, applying a conflict penalty while preserving discarded values in provenance. If the winning value remains below a confidence threshold, the field is left empty instead of emitting uncertain data. Multi-value fields (skills, phones, projects, experience, etc.) are normalized, unioned, and deduplicated.

Independent agreement across multiple sources increases confidence using a **noisy-OR** model rather than simple averaging. Overall profile confidence is computed separately from field-level merge logic.

Runtime Output Configuration

The canonical schema remains stable while a runtime configuration controls output shape. It can:

- select or rename fields,
- choose missing-value behavior (`null`, `omit`, or `error`),
- optionally expose confidence and provenance.

The same configuration generates the JSON Schema used to validate the final projected output.

Edge Cases & Scope

Handled

- Missing, malformed, or partial sources degrade gracefully.
- Multi-row CSVs produce one merged profile per candidate.
- Duplicate names with different email/phone anchors remain separate.
- Region-aware validation for bare phone numbers in notes.
- Public GitHub email is retained when available.

Out of Scope

- Reliable extraction of education/work history from free-text notes (requires NER/LLMs).
- Resume PDF/DOCX parsing.
- LinkedIn scraping.
- Cross-batch entity resolution beyond a submitted candidate bundle.